- Increasing the frequency of hard (cascading carry one) examples in the training data so the model has more hard examples to learn from. This improved training speed but did not reduce loss.
- Increasing the number of attention heads from 3 to 4 or 5 (while still using 1 layer) to provide more computing power.
- Changing the question format from "12345+22222=" to "12345+22222equals" giving the model more prediction steps after the question is revealed before it needs to state the first answer digit.
- Inserting "+" in the answer format (e.g. "12345+22222=034567" becomes "12345+22222=+034567" had no impact on accuracy or the algorithm.
- With n_layers = 1 increasing the number of attention heads from 3 to 4.
- Changing the n_layers to 2 and n_heads to 2.

The smallest model shape that did reduce loss significantly was 2 layers with 3 attention heads.

G Appendix: Model Loss

The model defaults to batch size = 64. The loss function is simple:

- Per Digit Loss: For "per digit" graphs and analysis, for a given answer digit, the loss used is negative log likelihood.
- All Digits Loss: For "all answer digits" graphs and analysis, the loss used is the mean of the "per digit" loss across all the answer digits.

In our experimental models, the number of digits in the question varies from 5 to 15, the number of layers varies from 1 to 4, the number of heads varies from 3 to 4. Each experimental model's loss is detailed in Tabs. 5 and 6.

During training, the models use an AdamW optimizer with a learning rate (LR) of 0.00008, weight decay of 0.1 and betas of (0.9, 0.98). During the first 5th of training the LR is linearly increased from 0.01 * LR to LR to help stabilize training. During the remainder of training, cosine annealing is used.

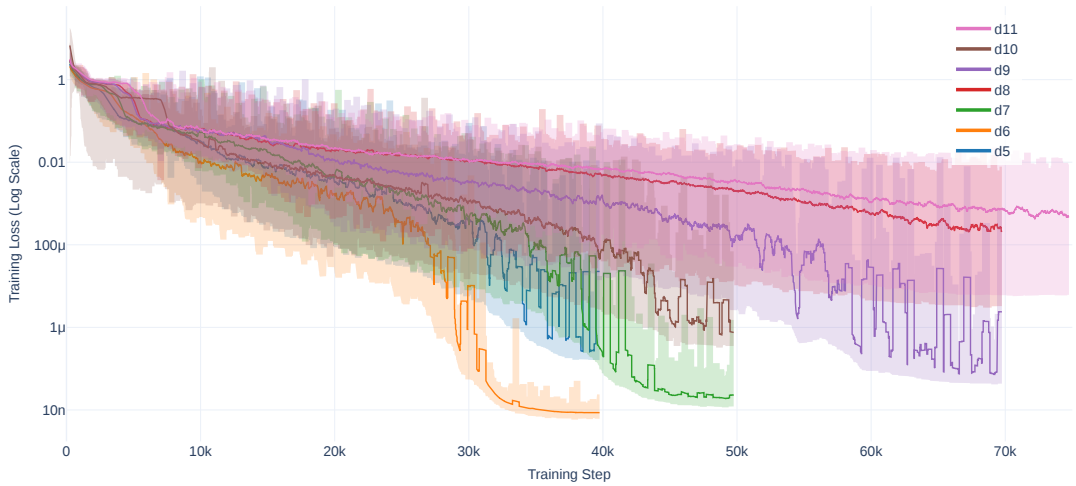

Figure 9: The 5- to 8-digit initialized mixed models have very low loss and >99.999% accuracy. With more digits the model loss and accuracy is worse. Details in Tab.6.

H Appendix: Experimental models

Forty-nine models were trained and analyzed (Refer Tabs. 5 and 6). The models and output of analysis are available on HuggingFace at `https://huggingface.co/PhilipQuirke` in the `quanta_maths_*` collections to support further research in AI Safety.

Table 5: Addition-only models studied. The number of addition failures per million questions is shown. The number of useful attention heads at token position and useful MLP layers at token position are shown - summarizing the data shown in figures like Fig 11.

Digits	Layers	Heads	Train Steps	Train Seed	Train loss	Add Fails / M	Heads used	MLPs used	Clopper-Pearson 95% Interval
5	1	3	30K	372001	9.4e-2	12621	15	6	[1.24e-2, 1.28e-2]
5	2	3	15K	372001	1.6e-8	0	30	16	[0.00e+0, 3.69e-6]
5	2	3	40K	372001	2.0e-9	0	22	15	[0.00e+0, 3.69e-6]
6	2	3	15K	372001	1.7e-8	2	31	17	[2.42e-7, 7.22e-6]
6	2	3	20K	173289	1.5e-8	0	28	17	[0.00e+0, 3.69e-6]
6	2	3	20K	572091	7.0e-9	0	35	17	[0.00e+0, 3.69e-6]
6	2	3	40K	372001	2.0e-9	0	29	17	[0.00e+0, 3.69e-6]
7	2	3	45K	173289	3.0e-9	0	31	20	[0.00e+0, 3.69e-6]
8	2	3	45K	173289	3.0e-9	0	35	22	[0.00e+0, 3.69e-6]
9	2	3	45K	173289	3.0e-9	0	54	27	[0.00e+0, 3.69e-6]
10	2	3	40K	572091	7.0e-9	0	44	28	[0.00e+0, 3.69e-6]
11	2	3	50K	173289	8.0e-9	2	56	29	[2.42e-7, 7.22e-6]
12	2	3	50K	173289	5.0e-9	3	50	33	[4.25e-7, 7.88e-6]
13	2	3	50K	173289	6.3e-8	1	66	31	[2.53e-8, 5.57e-6]
14	2	3	60K	173289	5.5e-6	199	68	35	[1.74e-4, 2.00e-4]
15	2	3	80K	572091	8.6e-8	10	93	58	[4.71e-6, 1.67e-5]

For each model the 'QuantaMathsTrain' Colab notebook generates two files:

- A "model.pth" file containing the model weights
- A "training_loss.json" file containing configuration information and training loss data

While, for each model the 'QuantaMathsAnalysis' Colab notebook generates two more files:

- A "behaviors.json" file containing generic "behavior" facts learnt about the model by the Colab e.g. P18L0H0 attends to tokens D3 and D'3
- A "features.json" file containing maths-specific "feature" facts learnt about the model by the Colab e.g. P18L0H0 performs the SC3 subtask.

I Appendix: Complexity

To analyze question difficulty, we categorized addition questions by the complexity of the computation required to solve the question, as shown in Tab. 8. The categories are arranged according to the number of digits that a carry bit has to cascade through.

J Appendix: N-Digit Addition

The addition models perform addition accurately. Visualizations that provided insights into the behavior of the model, aiding our interpretation of the algorithm, are below:

Some notes about the models:

- The models selected different attention heads in the early positions to use to do the same logical calculations.
- Some models use 2 attention heads per digit to do the SA calculation, whereas some models only uses one (and so are more compact).
- The PCA trigrams have difference appearances in different models (but the same interpretable clusters). Refer Figures 7
- Per answer digit, some models use the SC calculation, whereas some models optimize it out and rely solely on the ST value (and so are more compact).

Table 6: Subtraction-only and mixed models studied. The number of addition and subtraction failures per million questions is shown. The number of useful attention heads at token position and useful MLP layers at token position are shown - summarizing the data shown in figures like Fig 11.

Digits	Layers	Heads	Train Steps	Train Seed	Train loss	Add Fails / M	Sub Fails / M	Heads used	MLPs used	Clopper-Pearson 95% Interval
Subtraction models										
5	2	3	30K	372001	1.0e-3	N/A	3689	57	20	[3.57e-3, 3.81e-3]
6	2	3	30K	372001	5.8e-6	N/A	2	37	21	[2.42e-7, 7.22e-6]
6	2	3	30K	572091	5.8e-4	N/A	3889	58	21	[3.77e-3, 4.01e-3]
8	2	3	50K	173289	4.0e-9	N/A	1	65	26	[2.53e-8, 5.57e-6]
8	2	3	50K	371793	2.5e-5	N/A	487	71	28	[4.45e-4, 5.32e-4]
10	2	3	75K	173289	2.0e-3	N/A	6672	101	37	[6.50e-3, 6.72e-3]
12	2	3	75K	371793	3.4e-4	N/A	2175	96	32	[2.08e-3, 2.25e-3]
Mixed models										
5	3	4	40K	372001	9.0e-9	0	0	45	21	[0.00e+0, 3.69e-6]
6	3	4	40K	372001	5.0e-9	1	0	54	26	[2.53e-8, 5.57e-6]
7	3	4	50K	372001	2.0e-8	2	6	108	40	[3.45e-6, 1.58e-5]
8	3	4	60K	173289	4.7e-8	0	7	123	45	[2.81e-6, 1.44e-5
9	3	4	60K	173289	3.2e-7	1	33	140	46	[2.35e-5, 4.75e-5]
10	3	4	75K	173289	1.1e-6	2	295	143	53	[2.65e-4, 3.53e-4]
11	3	4	80K	572091	3.9e-8	0	13	138	50	[6.52e-6, 2.33e-5]
12	3	4	85K	572091	1.7e-8	2	10	167	55	[8.68e-6, 2.72e-5]
13	3	4	85K	572091	9.5e-6	399	4164	197	64	[4.41e-3, 4.70e-3]
Mixed models initialized with addition model										
5	2	3	40K	572091	2.5e-7	1	52	49	20	[5.35e-5, 8.48e-5]
6	2	3	40K	572091	2.4e-8	0	5	57	21	[1.63e-6, 1.10e-5]
6	3	3	40K	572091	1.8e-8	0	3	70	35	[4.25e-7, 7.88e-6]
6	3	3	80K	572091	1.6e-8	0	3	75	35	[4.25e-7, 7.88e-6]
6	3	4	40K	372001	8.0e-9	0	0	72	26	[0.00e+0, 3.69e-6]
6	3	4	40K	173289	1.4e-8	3	2	60	29	[2.16e-6, 1.18e-5]
6	3	4	50K	572091	2.9e-8	0	4	79	29	[8.11e-7, 9.28e-6]
7	3	4	50K	572091	1.8e-8	4	1	104	38	[1.03e-6, 1.08e-5]
8	3	4	70K	572091	4.3e-5	50	1196	116	42	[1.11e-3, 1.23e-3]
9	3	4	70K	572091	5.4e-8	1	4	160	50	[8.11e-7, 9.28e-6]
10	3	3	50K	572091	6.3e-7	6	7	90	45	[5.64e-6, 2.15e-5]
11	3	4	75K	572091	5.9e-5	11066	1120	141	47	[1.19e-2, 1.22e-2]
Mixed models initialized with add model. Reset useful heads every 100 steps										
6	4	4	40K	372001	1.7e-8	3	8	51	30	[5.64e-6, 2.15e-5]
Mixed models inited with add model. Reset useful hds & MLPs every 100 steps										
6	4	3	40K	372001	3.0e-4	17	3120	115	53	[3.06e-3, 3.30e-3]

K Appendix: Mixed Model Initialization

We experimented with three approaches to re-using the trained addition model in the "mixed" (addition and subtraction) model:

- Initialize Only: Initialize the untrained mixed model with the addition model weights before training begins.
- Freeze Attention: As per "Initialize Only", but also every 100 training steps recopy the attention head weights from the addition model into the partially-trained mixed model.
- Freeze All: As per "Initialize Only", but also every 100 training steps recopy the entire addition model (attention heads and MLP layers) into the partially-trained mixed model.

Our intuition was that "Initialize Only" would give the mixed model the most freedom to learn new algorithms, but that the "Freeze Attention" and "Freeze All" approaches would make the resulting trained mixed model easier to interpret (as we could reuse our addition model insights).